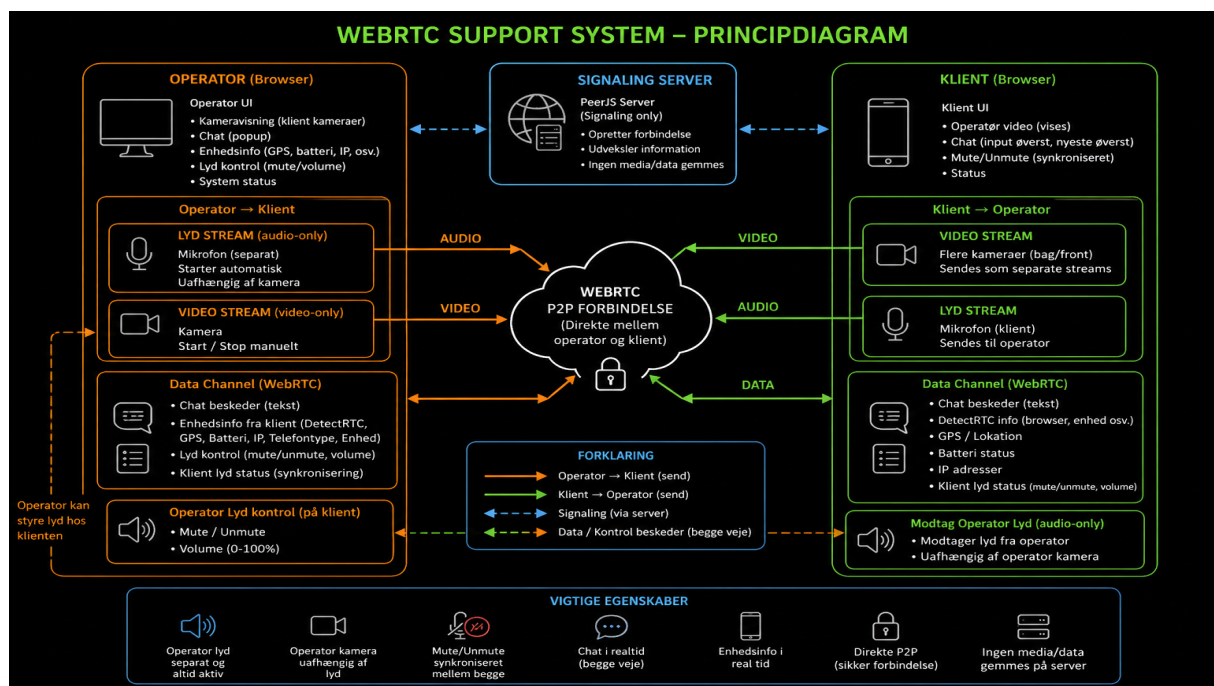
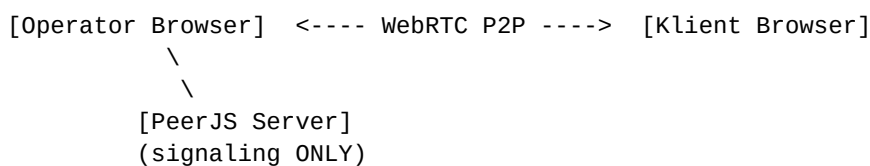


WebRTC Support System – Context Document (v1.0)

Formål

- Realtime support system mellem Operator (PC) og Klient (mobil)
- Live video (multi kamera fra klient)
- 2-vejs lyd (separate streams)
- Chat (messenger-style, nyeste øverst)
- Enhedsdata (GPS, batteri, IP osv.)
- Remote kontrol (lyd hos klient)

Arkitektur (princip)



Figur: Principskitse (forenklet)

Teknologier

- Frontend: HTML / CSS / Vanilla JS
- WebRTC API
- PeerJS
- Backend: Node.js + PeerJS server
- Deployment: Docker
- Kommunikation: WebRTC media streams + data channel

Streams

Operator → Klient

- Audio-only stream (type: operator-audio-only) – starter automatisk, uafhængig af kamera
- Video stream (type: operator-video-only) – startes manuelt

Klient → Operator

- Audio stream (type: client-audio) – altid aktiv
- Video streams (type: primitive-camera) – én stream pr kamera

Data Channel

- Chat
- GPS
- Batteri
- IP / netværk
- Enhedsinfo
- Audio kontrol
- Status sync

Lydsystem

Design

- Separate audio streams i begge retninger
- Uafhængig af video
- Synkroniserede mute-knapper

Flow

Operator mute
-> sender command
-> klient opdaterer
-> klient sender status
-> operator UI opdateres

Chat system

- Realtime via data channel
- Lokal historik (localStorage)
- Nyeste beskeder øverst

Klient UI

[Input]
[Nyeste besked]
[Ældre beskeder]

Operator UI

- Popup vindue

- Samme reversed struktur

Klient UI

- Auto-start (ingen knap)
- Video fra operator
- Chat under video
- Mute/unmute
- Minimal UI

Operator UI

- Kamera kontrol
- Lyd kontrol (lokal + remote)
- Chat popup
- System status
- Enhedsdata
- QR kode

Netværk

- PeerJS server (Docker)
- Port: 8443 (HTTPS)
- Kun signaling
- Media går direkte P2P

Sikkerhed

- HTTPS
- WebRTC encryption
- Ingen lagring af media
- Direkte forbindelser

Livscyklus

Klient

- Åbner link
- Auto-connect
- Starter kamera + lyd

Operator

- Åbner side
- Klar som operator
- Modtager streams

- Sender lyd automatisk

Deployment

```
docker build -t webrtc-support .
docker run -d \
  --name webrtc-support \
  --restart unless-stopped \
  -p 8443:8443 \
  webrtc-support
```

Version

- Version: 1.0
- Status: STABIL
- Backup: webrtc-support-v1.0

Version 2 idéer

- Reconnect / fail-safe
- Multi klient support
- Push-to-talk
- UI simplificering
- Logging

Quick Resume (til ny chat)

WebRTC support system:

- PeerJS + Docker
- Operator + klient
- Separate audio streams
- Multi kamera
- Chat reversed
- Remote audio control

Konklusion

- Realtime
- P2P
- Lav latency
- Stabilit (v1.0)
- Klar til videreudvikling